

Question			Marking Details	Marks Available																													
				AO1	AO2	AO3	TOTAL	Maths	Prac																								
6	(a)	i	445-450 <u>nm</u>		1		1																										
		ii	to absorb a greater range of wavelengths of light	1			1																										
	(b)	i	Can <u>only</u> {survive/ grow} in absence of oxygen/ ORA	1			1																										
		ii	B (1) Middle of the cell membrane is {non-polar/ made of fatty acids}/ fatty acids are {non-polar/ hydrophobic} (1) Only award 2 nd mark if 1 st correct		2		2																										
	(c)		Any four × (1) from: <table><tr><td></td><td>Sulfur bacteria</td><td>Green plants</td></tr><tr><td>A</td><td>1 photosystem</td><td>2 photosystems</td></tr><tr><td>B</td><td>H₂S used</td><td>H₂O used</td></tr><tr><td>C</td><td>Sulfur released</td><td>Oxygen released</td></tr><tr><td>D</td><td>{Bacteriochlorophyll/ carotenoids} in reaction centre</td><td>Chlorophyll a in reaction centre</td></tr><tr><td>E</td><td>Reduced NAD formed</td><td>reduced NADP formed</td></tr><tr><td>F</td><td>Protons pumped out of cell</td><td>Protons pumped into the thylakoid space</td></tr><tr><td>G</td><td>Cyclic photo-phosphorylation only</td><td></td></tr></table>		Sulfur bacteria	Green plants	A	1 photosystem	2 photosystems	B	H ₂ S used	H ₂ O used	C	Sulfur released	Oxygen released	D	{Bacteriochlorophyll/ carotenoids} in reaction centre	Chlorophyll a in reaction centre	E	Reduced NAD formed	reduced NADP formed	F	Protons pumped out of cell	Protons pumped into the thylakoid space	G	Cyclic photo-phosphorylation only			4		4		
	Sulfur bacteria	Green plants																															
A	1 photosystem	2 photosystems																															
B	H ₂ S used	H ₂ O used																															
C	Sulfur released	Oxygen released																															
D	{Bacteriochlorophyll/ carotenoids} in reaction centre	Chlorophyll a in reaction centre																															
E	Reduced NAD formed	reduced NADP formed																															
F	Protons pumped out of cell	Protons pumped into the thylakoid space																															
G	Cyclic photo-phosphorylation only																																
			QUESTION 6 TOTAL	2	7	0	9	0	0																								